

Lecture 9: Computer Networks

4.4 Basics of Computer Networks

Computer networks are like highways on which data travels. A computer network connects parts of a distributed system, including hardware and software. It enables sharing of data, devices, and resources—functions that are essential in today's world.

4.4.1 Computer Network and Networking

(i) Computer Network

A **computer network** is a group of computers and related equipment connected by communication links to share data and resources.

(ii) Networking

Networking is the act of joining computers and accessories to enable information exchange and resource sharing.

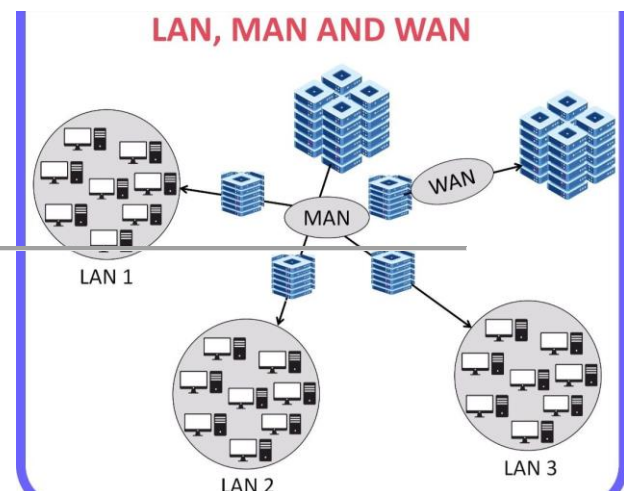
- Plays a vital role in **computers and telecommunications**.
 - Modern organizations rely on networking for **fast, inexpensive, and reliable communication**.
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4.4.2 Types of Computer Networks

Computer networks can be categorized by **size** and **purpose**. The size is defined by the geographical area they cover and the number of connected devices. Networks range from a few devices in a single room to millions across the globe.

Types of Networks

- **Local Area Network (LAN)**
- **Metropolitan Area Network (MAN)**
- **Wide Area Network (WAN)**



(i) Local Area Network (LAN)

- Covers a **limited area** such as schools, laboratories, homes, or office buildings.
 - Enables sharing of **internet access, storage space, and printers**.
 - Built with **inexpensive hardware** like hubs, switches, network adapters, and cables.
 - Facilitates sharing of **data and software**.
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(ii) Metropolitan Area Network (MAN)

- Spans an **entire city, college campus, or small region**.
 - Covers areas of **several miles** and may include multiple LANs.
 - Provides **fast communication** but is **expensive to establish**.
 - Typically used by **large businesses or universities**.
 - Requires **security measures** to prevent unauthorized access.
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(iii) Wide Area Network (WAN)

- Designed for **long-distance data transmission**.
 - Connects **cities, provinces, or countries**.
 - Uses **satellites, microwaves, or telecommunication links**.
 - Commonly used by **large businesses, research institutions, and educational organizations**.
 - May include multiple **MANs and LANs**.
 - Requires **expensive devices and dedicated connections**.
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Multiple Choice Questions (MCQs)

Section 4.4 Basics of Computer Networks

Q1. A computer network is often compared to:

- A) A library
- B) A highway
- C) A classroom
- D) A marketplace

Q2. A computer network connects:

- A) Only hardware
- B) Only software
- C) Both hardware and software
- D) None of these

Q3. Which of the following is NOT a resource shared in a computer network?

- A) Printers
- B) Internet connection
- C) File server
- D) Blackboard

Q4. Networking refers to:

- A) Connecting only printers
- B) Joining computers and accessories for sharing resources
- C) Installing software only
- D) None of the above

Q5. Networking plays a vital role in:

- A) Agriculture
- B) Telecommunication
- C) Cooking
- D) Sports

Section 4.4.2 Types of Computer Networks

Q6. Computer networks can be categorized by:

- A) Size and purpose
- B) Color and shape
- C) Speed and cost
- D) None of these

Q7. Which of the following is NOT a type of computer network?

- A) LAN
- B) MAN
- C) WAN
- D) PANCAKE

Q8. LAN stands for:

- A) Local Area Network
- B) Large Access Node
- C) Limited Access Network
- D) Local Application Network

Q9. LAN is generally used in:

- A) Entire countries
- B) Schools, homes, offices
- C) Satellites
- D) Oceans

Q10. Which hardware is commonly used to build LAN?

- A) Hubs and switches
 - B) Satellites
 - C) Microwaves
 - D) Fiber optics only
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Q11. MAN stands for:

- A) Metropolitan Area Network
- B) Medium Access Node
- C) Main Application Network
- D) Multiple Access Network

Q12. MAN usually covers:

- A) A single room
- B) A small region or city
- C) Entire countries
- D) Only homes

Q13. Which organizations typically set up MAN?

- A) Small shops
- B) Large businesses and universities
- C) Individual households
- D) Farmers

Q14. A disadvantage of MAN is:

- A) Slow communication
 - B) Expensive to establish
 - C) Cannot connect LANs
 - D) No security required
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Q15. WAN stands for:

- A) Wide Area Network
- B) Wireless Access Node
- C) World Application Network
- D) Wide Access Node

Q16. WAN is used for:

- A) Short-distance transmission
- B) Long-distance transmission
- C) Only gaming
- D) None of these

Q17. WAN connects:

- A) Only one city
- B) Cities, provinces, or countries
- C) Only homes
- D) Only offices

Q18. WAN technology may use:

- A) Satellites and microwaves
- B) Blackboard and chalk
- C) Books and pens
- D) None of these

Q19. WAN is usually set up by:

- A) Small households
- B) Large businesses, research, and educational organizations
- C) Farmers
- D) Sports clubs

Q20. WAN may include:

- A) Multiple LANs and MANs
 - B) Only one LAN
 - C) Only one MAN
 - D) None of these
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✓ Answer Key with Explanations

1. **B – Highway**
Networks are compared to highways where data travels.
2. **C – Both hardware and software**
Networks connect distributed systems including hardware and software.
3. **D – Blackboard**
Blackboard is not a network resource.
4. **B – Joining computers and accessories**
Networking enables exchange of information and sharing of resources.
5. **B – Telecommunication**
Networking is vital in computers and telecommunication fields.
6. **A – Size and purpose**
Networks are categorized by geographical size and purpose.
7. **D – PANCAKE**
PANCAKE is not a type of network.
8. **A – Local Area Network**
LAN stands for Local Area Network.
9. **B – Schools, homes, offices**
LAN is used in limited areas like schools, labs, homes, offices.
10. **A – Hubs and switches**
LAN uses inexpensive hardware like hubs, switches, adapters, cables.
11. **A – Metropolitan Area Network**
MAN stands for Metropolitan Area Network.
12. **B – A small region or city**
MAN covers cities, campuses, or small regions.
13. **B – Large businesses and universities**
MAN is expensive, so large organizations set it up.
14. **B – Expensive to establish**
MAN offers fast communication but is costly.
15. **A – Wide Area Network**
WAN stands for Wide Area Network.
16. **B – Long-distance transmission**
WAN is used for long-distance data transmission.
17. **B – Cities, provinces, or countries**
WAN connects large geographical areas.
18. **A – Satellites and microwaves**
WAN uses satellites, microwaves, and telecom links.
19. **B – Large businesses, research, and educational organizations**
WAN is used by organizations spread across long distances.
20. **A – Multiple LANs and MANs**
WAN may include multiple LANs and MANs.